

Study Mind AI Architecture

Core Components

User Interface (Streamlit)

PDF Uploads

Chatbot Interface

Study Artifacts

Orchestration (LangChain)

Document Loading

Chunking Tools

Prompt Formatting

Data Processing Pipeline

Ingestion (PyMuPDF)

Transformation (Recursive Splitting)

Vectorization (HuggingFace)

Storage (FAISS)

In-memory Vector Database

Semantic Search

Inference (Groq LLM)

RAG Prompt Execution

Ultra-low Latency

Module Breakdown

Core Engine

Vector Store Module

Prompts Module

Feature Processors

Study Tools

Analysis

Exam Prep

Conversational Layer

rag_chatbot.py

LangChain Memory

Scaling Strategy

Worker Nodes (Celery/Redis)

Stateless Vector Storage

SQL Persistence Layer